

of Questions: 30

Question #: 1

A 55-year-old woman comes to the physician because of a 2-month history of early satiety, nausea, and vomiting of partially digested food. She also has epigastric fullness, especially after meals, which lessens after vomiting. Imaging studies of the abdomen show small bowel obstruction secondary to an aneurysm of the superior mesenteric artery at its origin from the aorta. Which of the following is most likely the site of compression?

- A. Pyloric region of the stomach
 - B. 1st part of the duodenum
 - C. 3rd part of the duodenum
 - D. Terminal ileum
 - E. Ascending colon
-

Question #: 2

A 25-year-old man is brought to the emergency department because of a 2-hour history of severe abdominal pain. Physical examination shows rebound tenderness over the right lower quadrant and a positive Psoas sign. Ultrasonography of the abdomen shows an enlarged appendix with thickened walls. Laparoscopic surgical removal is difficult because there is a fat-filled peritoneal fold, preventing visualization of the appendix. Which of the following is the most likely embryological origin of the structure obscuring the view?

- A. Foregut
 - B. Dorsal mesentery
 - C. Ventral mesentery
 - D. Midgut
 - E. Hindgut
-

Question #: 3

A 70-year-old woman is brought to the emergency department because of a 4-hour history of intense epigastric abdominal pain that radiates to the back. Physical examination shows jaundice. Endoscopic retrograde pancreatography shows a tumor at the head of the pancreas. Which of the following nervous structures most likely carries the visceral afferents that were stimulated to result in the patient's pain?

- A. Vagus
 - B. Lesser splanchnics
 - C. Greater splanchnics
 - D. Pelvic splanchnics
 - E. Least splanchnics
-

Question #: 4

A 3-month-old boy is brought to the emergency department by his parents because of a 6-hour history of vomiting. Imaging studies of the abdomen show gastric outlet obstruction due to pyloric stenosis. Which of the following arterial blood gas results is most likely in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V

	pH	PCO₂	HCO₃⁻
I	Normal	Increased	Decreased
II	Increased	Decreased	Increased
III	Decreased	Increased	Normal
IV	Increased	Increased	Increased
V	Decreased	Decreased	Normal

Question #: 5

A 46-year-old woman comes to the physician because of a 2-week history of yellowing of her skin and eyes. She also has pale and bulky feces. Physical examination shows scleral icterus. Laboratory studies show elevated serum alkaline phosphatase levels. Abdominal ultrasonography shows dilation of the biliary tree and the presence of gall stones in the common bile duct. Which of the following sets of laboratory results will most likely be found in this patient?

- A. I
- B. II
- C. III
- D. IV
- E. V

	Type of serum bilirubin	Urine bilirubin	Urine urobilinogen
I	Unconjugated	Absent	Absent
II	Unconjugated	Present	Increased
III	Unconjugated	Absent	Increased
IV	Conjugated	Absent	Increased
V	Conjugated	Present	Absent

Question #: 6

A 20-year-old man comes to the physician because of a 1-day history of passing brown-colored urine, muscle pain and muscle cramps after running on the treadmill for about 30 minutes. He says he can tolerate mild exercise. Examination of muscle biopsy specimen shows excessive glycogen accumulation. The structure and branching pattern of glycogen is normal. Muscle triglyceride levels are normal. Examination of a liver biopsy specimen shows normal glycogen structure and content. Which of the following is the most likely diagnosis?

- A. McArdle disease
- B. Myopathic fatty acid oxidation defect
- C. Medium chain acyl CoA dehydrogenase deficiency
- D. vonGierke disease
- E. Duchenne muscular dystrophy

Question #: 7

An investigator is studying the regulation of acetyl CoA carboxylase. It is noted that the presence of citrate activates the enzyme. Which of the following changes to the enzyme is most likely observed under these conditions?

- A. Polymer formation
 - B. Phosphorylation
 - C. Dephosphorylation
 - D. Monomer formation
 - E. Up-regulation and increased synthesis
-

Question #: 8

A 9-year-old girl is brought to the physician by her parents because of a 2-week history of painful muscle cramps during exercise. Laboratory studies show normal plasma glucose level but no increase of lactate in the blood following exercise. Examination of a muscle biopsy specimen shows increased glycogen content with the normal pattern of branching in its structure. Inherited deficiency of which of the following is the most likely diagnosis?

- A. Carnitine palmitoyl transferase-II (CPT-II)
 - B. Glycogen debranching enzyme
 - C. Glycogen branching enzyme
 - D. Muscle glycogen phosphorylase
 - E. Pyruvate dehydrogenase complex
 - F. Medium chain acyl CoA dehydrogenase (MCAD)
-

Question #: 9

A 50-year-old man comes to the physician because of a 3-week history of yellowing of the eyes. Physical examination shows jaundice and ascites. Results of laboratory studies are shown. Which of the following organs are also most likely affected in this disorder?

- A. Lungs
- B. Basal ganglia, cornea
- C. Pancreas, skin
- D. Blood vessels and gums
- E. Twisted hair, blood vessel aneurysms
- F. Pancreas and lungs

	Patient	Normal range
Alanine aminotransferase (ALT)	180	40-60
Aspartate aminotransferase (AST)	250	40-60
Alkaline phosphatase (ALP)	140	50-70
Gamma glutamyl transferase (GGT)	130	50-70
Serum ferritin	340	20-250 ng/mL

Question #: 10

A 32-year-old woman is brought to the emergency department by her roommate because of a 2-hour history of confusion and vomiting. She has a history of attempted suicide in the previous 6 months. Physical examination shows abdominal tenderness. Gastric lavage analysis shows an overdose of acetaminophen. Laboratory studies show abnormal liver function. Which of the following laboratory results is most likely observed in this patient?

- A. Elevated blood urea levels
- B. Elevated serum alpha fetoprotein levels
- C. Elevated serum alanine aminotransferase levels
- D. Decreased serum alkaline phosphatase levels
- E. Serum AST: ALT ratio of 2:1

Question #: 11

Six miners are admitted to the hospital after being trapped in a collapsed mine for one week with only water and no food. Laboratory studies show an average fasting plasma glucose of 70 mg/dL. Which of the following metabolic pathways is most likely responsible for the maintenance of the plasma glucose levels in these patients?

- A. Triacylglycerol hydrolysis in adipose tissue
- B. Glycogenolysis in the liver
- C. Ketone body synthesis in the liver

- D. Gluconeogenesis in the liver
 - E. Glycolysis in erythrocytes
-

Question #: 12

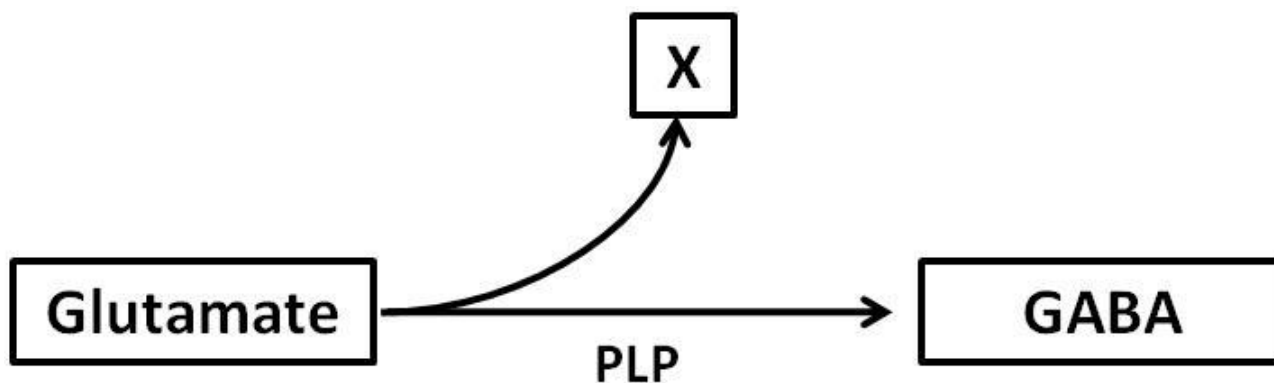
A 25-day-old boy is brought to the emergency department by his mother because of a 2-day history of drowsiness and progressively increasing unresponsiveness. Laboratory studies show respiratory alkalosis and a deficiency of carbamoyl phosphate synthetase-I (CPS-I). Which of the following additional laboratory findings is most likely in this patient?

- A. Elevated orotic acid in urine
 - B. Elevated arginino succinic acid levels
 - C. Low plasma glutamine levels
 - D. Elevated blood ammonia levels
 - E. Elevated urinary C-8 and C10 acyl carnitine
 - F. Elevated blood urea levels
 - G. Elevated homocysteine levels
-

Question #: 13

An investigator is studying the formation of a specific inhibitory neurotransmitter shown in the figure. Which of the following best describes the type of reaction shown in the figure?

- A. Carboxylation
- B. Transamination
- C. Decarboxylation
- D. Dehydrogenation
- E. Oxidative deamination



Question #: 14

A 10-day-old female newborn is brought to the physician by her mother for a follow-up examination because of a 1-week history of yellowing of the skin. Laboratory studies initially showed total bilirubin levels of 4mg/dL (Normal: 0.1-1 mg/dL) and direct bilirubin was not detected. Today, her laboratory studies show normal bilirubin levels. Which of the following is the best explanation for her symptoms?

- A. A temporary deficiency of heme oxygenase
- B. A low activity of UDP-glucuronyl transferase
- C. Dubin-Johnson syndrome
- D. Crigler-Najjar syndrome type-I
- E. An abnormally low amount of serum albumin

Question #: 15

A 2-year-old girl is brought to the emergency department by her parents because of a sudden-onset of seizures. She has had similar incidents over the previous six months. Results of laboratory studies on admission are shown in the table. Which of the following inherited enzyme deficiency disorders is most likely in this patient?

- A. Glucose 6-phosphatase
- B. Muscle glycogen phosphorylase
- C. Medium chain acyl CoA dehydrogenase deficiency

- D. Pyruvate dehydrogenase complex deficiency
E. Aldolase B

	Patient results	Reference ranges
pH	7.3	7.36-7.42
HCO ₃ ⁻	20 mEq/L	24-26 mEq/L
Fasting blood glucose	50 mg/dL	70-90 mg/dL
Serum lactate	Increased	-NA-

Question #: 16

An investigator studying muscle metabolic changes during extreme exertion and strenuous activity observes high levels of intracellular AMP. This results in the allosteric activation of various skeletal muscle enzymes. In which of the following skeletal muscle enzymes is this effect most likely observed?

- A. Phosphofructokinase-1 and glycogen phosphorylase
- B. Hexokinase and glycogen phosphorylase
- C. Glycogen synthase and glycogen phosphorylase
- D. Debranching enzyme and pyruvate kinase
- E. Lactate dehydrogenase and phosphofructokinase-1
- F. Phosphofructokinase-1 and glycogen phosphorylase kinase

Question #: 17

A 35-year-old woman comes to the physician because of a 2-week history of easy bruising. Four months ago, she was diagnosed with gall stones 4 months ago. Stool examination shows steatorrhea. Laboratory studies show increased direct bilirubin levels and prolonged prothrombin time. Which of the following enzyme activities is most likely impaired in this patient?

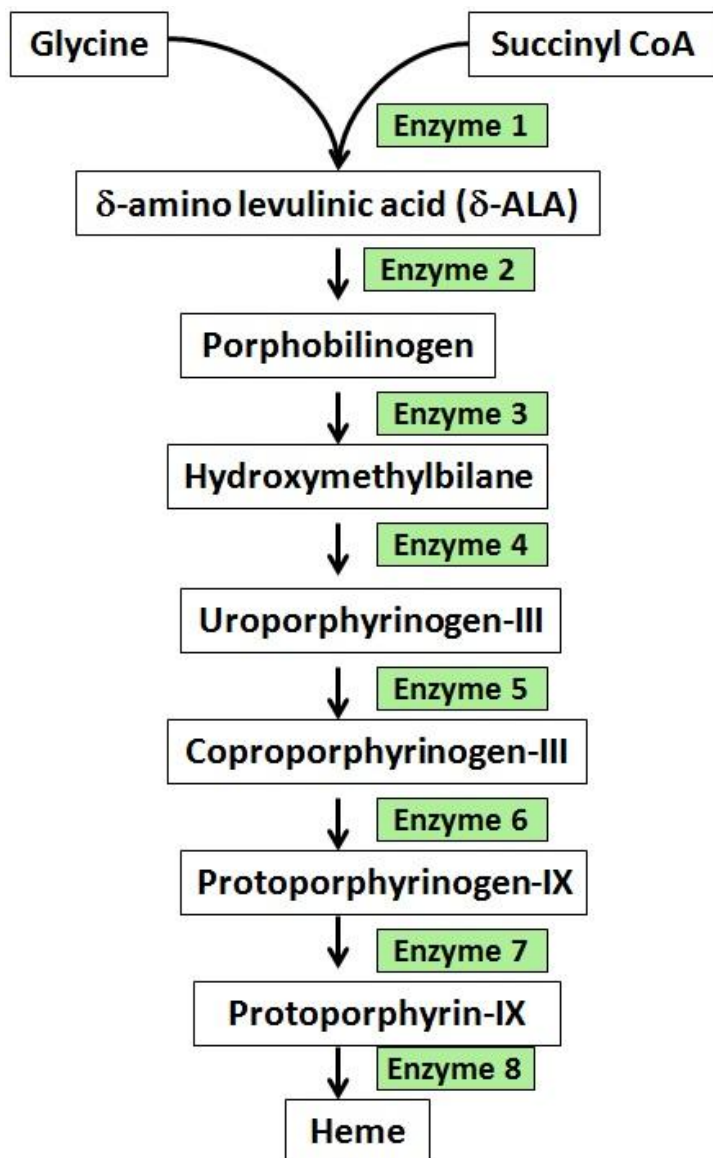
- A. Methylmalonyl CoA mutase
- B. Propionyl CoA carboxylase
- C. γ -glutamyl carboxylase in the liver

- D. 25-hydroxylase in the liver
 - E. 1-hydroxylase in the kidney
 - F. Prolyl hydroxylase
-

Question #: 18

An investigator is studying the pathway of heme synthesis shown in the figure in the liver and bone marrow. Which of the following best describes this pathway?

- A. Enzyme 6 is inhibited by lead
- B. Enzyme 1 requires PLP as a coenzyme
- C. Inherited deficiency of Enzyme 3 results in photosensitivity
- D. Enzyme 5 deficiency causes congenital erythropoietic porphyria
- E. Inherited deficiency of Enzyme 4 causes neuropsychiatric manifestations



Question #: 19

A 24-year-old woman had a breakfast of cereal, coffee with sugar, toast, fried bacon and eggs. Liver metabolism is observed one hour later. Which of the following metabolic effects is most likely observed in the hepatocytes?

- A. Increased glycogen phosphorylase activity
- B. Increased acetyl CoA carboxylase activity
- C. Decreased phosphofructokinase-1 activity
- D. Increased carnitine palmitoyl transferase-I activity
- E. Increased pyruvate carboxylase activity

F. Increased mitochondrial HMG-CoA synthase activity

Question #: 20

An investigator is studying metabolism in experimental rats under different metabolic conditions. When fed a diet rich in carbohydrates, large amounts of citrate are formed in the cytosol of hepatocytes of experimental animals. Which of the following is the most likely effect of this intermediate on lipid metabolism?

- A. Activation of the carnitine shuttle system
 - B. Activation of fatty acid synthesis
 - C. Inhibition of carnitine palmitoyl transferase-I (CPT-I)
 - D. Inhibition of cholesterol biosynthesis
 - E. Activation of bile acid synthesis
 - F. Activation of ketogenesis
-

Question #: 21

A 24-year-old man fasts for 2 days because he wants to reduce his weight for an upcoming modeling show. He takes only sips of orange juice equal to about 150 ml per day. Which of the following metabolic changes would most likely be present in the liver in this individual?

- A. Dephosphorylation of the PFK-2/FBPase-2 bifunctional enzyme
 - B. Induction of phosphoenolpyruvate carboxykinase
 - C. Dephosphorylation of glycogen phosphorylase kinase
 - D. Inactivation of protein kinase A
 - E. Dephosphorylation of glycogen phosphorylase
 - F. Increased levels of fructose 2,6 bisphosphate
 - G. Increased activity of glycogen synthase
-

Question #: 22

A 36-year-old woman comes to the physician because of a 2-month history of intermittent abdominal pain. Physical examination shows tenderness over the right

upper quadrant of the abdomen. Results of laboratory studies are shown. Which of the following is the most likely diagnosis?

- A. Hemolytic anemia
- B. Acute viral hepatitis
- C. Chronic alcoholic cirrhosis
- D. Gall stones in the common bile duct
- E. Crigler Najjar syndrome type-I
- F. Gilbert syndrome

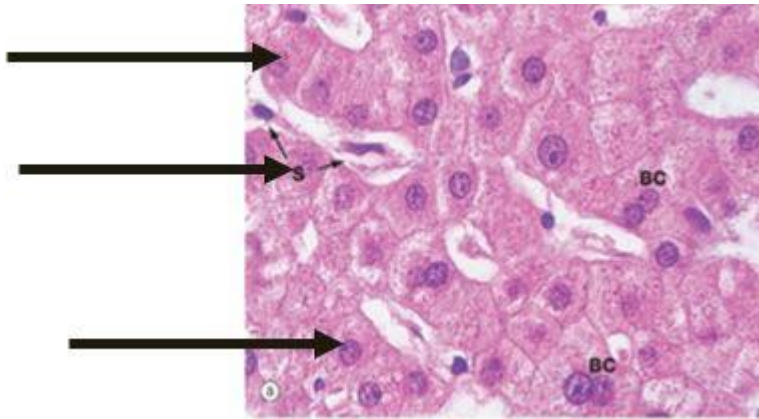
	Patient results	Reference results
Total serum bilirubin	6	1-2 mg/dL
Direct serum bilirubin	5.5	<0.2 mg/dL
Urine bilirubin	Present	Absent
Urine urobilinogen	Absent	Present
Fecal stercobilin	Absent	Present

Question #: 23

A 33-year-old woman comes to the physician because of a 1-week history of loss of appetite, nausea, weakness, and dark urine. Her temperature is 38°C (101°F), pulse is 100/min and respirations are 22/min. Physical examination shows right upper quadrant abdominal tenderness. Laboratory studies show antibodies against hepatitis B virus. A diagnosis of acute hepatitis B infection is made. In this condition, there is significant inflammation of the cells shown at the tips of the arrows in the attached photomicrograph. Which of the following is the most likely function of these cells?

- A. Production of bilirubin
- B. Phagocytosis of red blood cells
- C. Storage of vitamin A as retinyl esters
- D. Concentration of bile

E. Synthesis of secretin



Question #: 24

A research group is investigating the role of bacteria in inflammatory bowel diseases of the gastrointestinal tract. They have identified a receptor that binds Type I pili on a subset of Gram-negative enterobacilli such as *Escherichia coli*. The group hypothesizes that the receptor is important for immunosurveillance at the mucosal surface and the initiation of an effective adaptive immune response. It appears that the receptor may be a specific marker for a particular cell type. Which of the following best characterizes the cell type most likely involved in this function?

- A. Acidophilic apical secretory vesicles
- B. Columnar cells with a striated border
- C. Produces hormones that act on the gallbladder and pancreas
- D. Large accumulations of mucinogen granules in the apical cytoplasm
- E. Abundant microfolds on the apical surface

Question #: 25

A 56-year-old man comes to the physician because of a 2-week history of heartburn. Physical examination shows epigastric tenderness. Laboratory studies show no abnormalities. An endoscopic exam is requested, and multiple biopsies are taken during this procedure. One of the light micrographs of the biopsy specimen is attached for your reference. Which of the following histological features helps to identify the segment where the biopsy was taken from?

- A. Intestinal crypts of Lieberkühn
- B. Esophageal glands
- C. Von Ebner's glands
- D. Enteroendocrine glands
- E. Brunner's glands



Question #: 26

A 52-year-old woman comes to the physician because of a 3-day history of diarrhea. She has been drinking only small amounts because she feels nauseous. Physical examination shows dry lips and increased capillary refill times. She is found to have hypovolemia and needed fluid replacement. She receives an oral rehydration solution containing NaCl, glucose, KCl and citrate dissolved in water. Which of the following intestinal uptake mechanisms is most likely targeted by this rehydration solution to restore fluid in the patient?

- A. Sodium-chloride co-transport
 - B. Passive sodium absorption
 - C. Chloride absorption via CFTR channels
 - D. Sodium-glucose co-transport
 - E. Active potassium absorption
-

Question #: 27

A 3-week-old boy is brought to the physician by his mother for a routine examination. The mother says that his bowel movements increase after every feeding. Physical examination shows no abnormalities. The physician reassures the mother not to worry as this is physiological. Which of the following reflexes best explains the cause of this physiological occurrence in the newborn?

- A. Gastroileal
 - B. Gastrocolic
 - C. Intestino-intestinal
 - D. Defecation
 - E. Peristaltic rush
-

Question #: 28

A 3-year-old girl is brought to the physician by her mother because of a 2-week history of diarrhea characterized by pale, bulky and foul-smelling stools. Her mother is also concerned about her failure to thrive. Laboratory studies show elevated sweat chloride concentrations. Genetic studies show CFTR gene mutation. Decreased pancreatic secretion of which of the following best explains diarrhea in this patient?

- A. Lipase
- B. Secretin
- C. Amylase
- D. Cholecystokinin
- E. Bicarbonate

Question #: 29

A 28-year-old man comes to the clinic because of chronic diarrhea. He says his daily stool output is approximately 2 liters (tenfold increase). He has just recently undergone a surgical resection of his terminal ileum because of Crohn's disease. Which of the following substances are most likely responsible for his current condition?

- A. Bile acids
- B. Inflammatory cytokines
- C. Short chain fatty acids
- D. Prostaglandins
- E. Vasoactive intestinal polypeptides

Question #: 30

A 45-year-old woman comes to the physician because of a 2-month history of fatigue, weight loss, bone pains and diarrhea. She says she passes a stool that is sticky and foul-smelling. She has a 10-year history of Crohn's disease and underwent an ileal resection 3 months ago. Her medical records show that 60 cm of distal ileum was removed during the surgery. Absorption of which of the following substances most likely are affected following the surgery in this patient?

- A. Vitamin B12 and calcium
- B. Vitamin B12 and bile acids
- C. Calcium and fat-soluble vitamins
- D. Fat-soluble vitamins and glucose
- E. Bile acids and sodium